MSG No. 67.509 C_amma [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `mmm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ 0\right]$	`[1,6,12,15,19,24,26,29]`
2	$\left[0,\ \frac{3}{4},\ 0\right]$	`[2,5,11,16,20,23,25,30]`
3	$\left[\frac{1}{2},\ \frac{1}{4},\ 0\right]$	`[3,8,10,13,17,22,28,31]`
4	$\left[\frac{1}{2},\ \frac{3}{4},\ 0\right]$	`[4,7,9,14,18,21,27,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `mm'm'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,5,6,19,20,23,24]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[3,4,7,8,17,18,21,22]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,10,13,14,27,28,31,32]`
4	$\left[0,\ \frac{1}{2},\ 0\right]$	`[11,12,15,16,25,26,29,30]`

Table 3: Wyckoff site: `4c`, site symmetry: `mmm'`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ \frac{1}{2}\right]$	`[1,6,12,15,19,24,26,29]`
2	$\left[0,\ \frac{3}{4},\ \frac{1}{2}\right]$	`[2,5,11,16,20,23,25,30]`
3	$\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[3,8,10,13,17,22,28,31]`
4	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[4,7,9,14,18,21,27,32]`

Table 4: Wyckoff site: `4d`, site symmetry: `mm'm'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,5,6,19,20,23,24]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[3,4,7,8,17,18,21,22]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,13,14,27,28,31,32]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[11,12,15,16,25,26,29,30]`

Table 5: Wyckoff site: `4e`, site symmetry: `m'mm'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ 0\right]$	`[1,3,13,15,22,24,26,28]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ 0\right]$	`[2,4,14,16,21,23,25,27]`
3	$\left[\frac{3}{4},\ \frac{3}{4},\ 0\right]$	`[5,7,9,11,18,20,30,32]`
4	$\left[\frac{3}{4},\ \frac{1}{4},\ 0\right]$	`[6,8,10,12,17,19,29,31]`

Table 6: Wyckoff site: `4f`, site symmetry: `m'm'm'`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, 0\right]$	`[1,2,3,4,21,22,23,24]`
2	$\left[\frac{3}{4}, 0, 0\right]$	`[5,6,7,8,17,18,19,20]`
3	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[9,10,11,12,29,30,31,32]`
4	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[13,14,15,16,25,26,27,28]`

Table 7: Wyckoff site: `4g`, site symmetry: `m'mm'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[1,3,13,15,22,24,26,28]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,4,14,16,21,23,25,27]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[5,7,9,11,18,20,30,32]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[6,8,10,12,17,19,29,31]`

Table 8: Wyckoff site: `4h`, site symmetry: `m'm'm'`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[1,2,3,4,21,22,23,24]`
2	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[5,6,7,8,17,18,19,20]`
3	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,10,11,12,29,30,31,32]`
4	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14,15,16,25,26,27,28]`

Table 9: Wyckoff site: `8i`, site symmetry: `m2'm'`

No.	position	mapping
1	$[0, y, 0]$	`[1,6,19,24]`
2	$[0, -y, 0]$	`[2,5,20,23]`
3	$\left[\frac{1}{2}, y, 0\right]$	`[3,8,17,22]`
4	$\left[\frac{1}{2}, -y, 0\right]$	`[4,7,18,21]`
5	$\left[\frac{1}{2}, y + \frac{1}{2}, 0\right]$	`[9,14,27,32]`
6	$\left[\frac{1}{2}, \frac{1}{2} - y, 0\right]$	`[10,13,28,31]`
7	$\left[0, y + \frac{1}{2}, 0\right]$	`[11,16,25,30]`
8	$\left[0, \frac{1}{2} - y, 0\right]$	`[12,15,26,29]`

Table 10: Wyckoff site: `8j`, site symmetry: `m2'm'`

No.	position	mapping
1	$\left[0, y, \frac{1}{2}\right]$	`[1,6,19,24]`
2	$\left[0, -y, \frac{1}{2}\right]$	`[2,5,20,23]`
3	$\left[\frac{1}{2}, y, \frac{1}{2}\right]$	`[3,8,17,22]`
4	$\left[\frac{1}{2}, -y, \frac{1}{2}\right]$	`[4,7,18,21]`
5	$\left[\frac{1}{2}, y + \frac{1}{2}, \frac{1}{2}\right]$	`[9,14,27,32]`
6	$\left[\frac{1}{2}, \frac{1}{2} - y, \frac{1}{2}\right]$	`[10,13,28,31]`

continued ...

Table 10

No.	position	mapping
7	$\left[0,\, y+\frac{1}{2},\, \frac{1}{2}\right]$	`[11,16,25,30]`
8	$\left[0,\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	`[12,15,26,29]`

Table 11: Wyckoff site: `8k`, site symmetry: `m'2m'`

No.	position	mapping
1	$\left[\frac{1}{4},\, y,\, 0\right]$	`[1,3,22,24]`
2	$\left[\frac{1}{4},\, -y,\, 0\right]$	`[2,4,21,23]`
3	$\left[\frac{3}{4},\, -y,\, 0\right]$	`[5,7,18,20]`
4	$\left[\frac{3}{4},\, y,\, 0\right]$	`[6,8,17,19]`
5	$\left[\frac{3}{4},\, y+\frac{1}{2},\, 0\right]$	`[9,11,30,32]`
6	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, 0\right]$	`[10,12,29,31]`
7	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, 0\right]$	`[13,15,26,28]`
8	$\left[\frac{1}{4},\, y+\frac{1}{2},\, 0\right]$	`[14,16,25,27]`

Table 12: Wyckoff site: `8l`, site symmetry: `m'2m'`

No.	position	mapping
1	$\left[\frac{1}{4},\, y,\, \frac{1}{2}\right]$	`[1,3,22,24]`
2	$\left[\frac{1}{4},\, -y,\, \frac{1}{2}\right]$	`[2,4,21,23]`
3	$\left[\frac{3}{4},\, -y,\, \frac{1}{2}\right]$	`[5,7,18,20]`
4	$\left[\frac{3}{4},\, y,\, \frac{1}{2}\right]$	`[6,8,17,19]`
5	$\left[\frac{3}{4},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	`[9,11,30,32]`
6	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	`[10,12,29,31]`
7	$\left[\frac{1}{4},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	`[13,15,26,28]`
8	$\left[\frac{1}{4},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	`[14,16,25,27]`

Table 13: Wyckoff site: `8m`, site symmetry: `2'mm'`

No.	position	mapping
1	$\left[x,\, \frac{1}{4},\, 0\right]$	`[1,15,24,26]`
2	$\left[x,\, \frac{3}{4},\, 0\right]$	`[2,16,23,25]`
3	$\left[\frac{1}{2}-x,\, \frac{1}{4},\, 0\right]$	`[3,13,22,28]`
4	$\left[\frac{1}{2}-x,\, \frac{3}{4},\, 0\right]$	`[4,14,21,27]`
5	$\left[-x,\, \frac{3}{4},\, 0\right]$	`[5,11,20,30]`
6	$\left[-x,\, \frac{1}{4},\, 0\right]$	`[6,12,19,29]`
7	$\left[x+\frac{1}{2},\, \frac{3}{4},\, 0\right]$	`[7,9,18,32]`
8	$\left[x+\frac{1}{2},\, \frac{1}{4},\, 0\right]$	`[8,10,17,31]`

Table 14: Wyckoff site: `8n`, site symmetry: `2'mm'`

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$	`[1,15,24,26]`
2	$\left[x,\ \frac{3}{4},\ \frac{1}{2}\right]$	`[2,16,23,25]`
3	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$	`[3,13,22,28]`
4	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	`[4,14,21,27]`
5	$\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$	`[5,11,20,30]`
6	$\left[-x,\ \frac{1}{4},\ \frac{1}{2}\right]$	`[6,12,19,29]`
7	$\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[7,9,18,32]`
8	$\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[8,10,17,31]`

Table 15: Wyckoff site: `8o`, site symmetry: `2m'm'`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,2,23,24]`
2	$\left[\frac{1}{2}-x,\ 0,\ 0\right]$	`[3,4,21,22]`
3	$[-x,\ 0,\ 0]$	`[5,6,19,20]`
4	$\left[x+\frac{1}{2},\ 0,\ 0\right]$	`[7,8,17,18]`
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,10,31,32]`
6	$\left[-x,\ \frac{1}{2},\ 0\right]$	`[11,12,29,30]`
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ 0\right]$	`[13,14,27,28]`
8	$\left[x,\ \frac{1}{2},\ 0\right]$	`[15,16,25,26]`

Table 16: Wyckoff site: `8p`, site symmetry: `2m'm'`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{2}\right]$	`[1,2,23,24]`
2	$\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$	`[3,4,21,22]`
3	$\left[-x,\ 0,\ \frac{1}{2}\right]$	`[5,6,19,20]`
4	$\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[7,8,17,18]`
5	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,31,32]`
6	$\left[-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[11,12,29,30]`
7	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[13,14,27,28]`
8	$\left[x,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[15,16,25,26]`

Table 17: Wyckoff site: `8q`, site symmetry: `mm2`

No.	position	mapping
1	$\left[0,\ \frac{1}{4},\ z\right]$	`[1,6,12,15]`
2	$\left[0,\ \frac{3}{4},\ -z\right]$	`[2,5,11,16]`
3	$\left[\frac{1}{2},\ \frac{1}{4},\ -z\right]$	`[3,8,10,13]`
4	$\left[\frac{1}{2},\ \frac{3}{4},\ z\right]$	`[4,7,9,14]`
5	$\left[\frac{1}{2},\ \frac{1}{4},\ z\right]$	`[17,22,28,31]`
6	$\left[\frac{1}{2},\ \frac{3}{4},\ -z\right]$	`[18,21,27,32]`
7	$\left[0,\ \frac{1}{4},\ -z\right]$	`[19,24,26,29]`

continued ...

Table 17

No.	position	mapping
8	$\left[0,\ \frac{3}{4},\ z\right]$	[20,23,25,30]

Table 18: Wyckoff site: 8r, site symmetry: m'm2'

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	[1,15,22,28]
2	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	[2,16,21,27]
3	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	[3,13,24,26]
4	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	[4,14,23,25]
5	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	[5,11,18,32]
6	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	[6,12,17,31]
7	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	[7,9,20,30]
8	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	[8,10,19,29]

Table 19: Wyckoff site: 8s, site symmetry: mm'2'

No.	position	mapping
1	$[0,\ 0,\ z]$	[1,6,20,23]
2	$[0,\ 0,\ -z]$	[2,5,19,24]
3	$\left[\frac{1}{2},\ 0,\ -z\right]$	[3,8,18,21]
4	$\left[\frac{1}{2},\ 0,\ z\right]$	[4,7,17,22]
5	$\left[\frac{1}{2},\ \frac{1}{2},\ z\right]$	[9,14,28,31]
6	$\left[\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[10,13,27,32]
7	$\left[0,\ \frac{1}{2},\ -z\right]$	[11,16,26,29]
8	$\left[0,\ \frac{1}{2},\ z\right]$	[12,15,25,30]

Table 20: Wyckoff site: 8t, site symmetry: m'm'2

No.	position	mapping
1	$\left[\frac{1}{4},\ 0,\ z\right]$	[1,4,22,23]
2	$\left[\frac{1}{4},\ 0,\ -z\right]$	[2,3,21,24]
3	$\left[\frac{3}{4},\ 0,\ -z\right]$	[5,8,18,19]
4	$\left[\frac{3}{4},\ 0,\ z\right]$	[6,7,17,20]
5	$\left[\frac{3}{4},\ \frac{1}{2},\ z\right]$	[9,12,30,31]
6	$\left[\frac{3}{4},\ \frac{1}{2},\ -z\right]$	[10,11,29,32]
7	$\left[\frac{1}{4},\ \frac{1}{2},\ -z\right]$	[13,16,26,27]
8	$\left[\frac{1}{4},\ \frac{1}{2},\ z\right]$	[14,15,25,28]

Table 21: Wyckoff site: 16u, site symmetry: .m.

No.	position	mapping
1	$\left[x,\ \frac{1}{4},\ z\right]$	[1,15]

continued ...

Table 21

No.	position	mapping
2	$\left[x,\ \frac{3}{4},\ -z\right]$	[2,16]
3	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ -z\right]$	[3,13]
4	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ z\right]$	[4,14]
5	$\left[-x,\ \frac{3}{4},\ -z\right]$	[5,11]
6	$\left[-x,\ \frac{1}{4},\ z\right]$	[6,12]
7	$\left[x+\frac{1}{2},\ \frac{3}{4},\ z\right]$	[7,9]
8	$\left[x+\frac{1}{2},\ \frac{1}{4},\ -z\right]$	[8,10]
9	$\left[x+\frac{1}{2},\ \frac{1}{4},\ z\right]$	[17,31]
10	$\left[x+\frac{1}{2},\ \frac{3}{4},\ -z\right]$	[18,32]
11	$\left[-x,\ \frac{1}{4},\ -z\right]$	[19,29]
12	$\left[-x,\ \frac{3}{4},\ z\right]$	[20,30]
13	$\left[\frac{1}{2}-x,\ \frac{3}{4},\ -z\right]$	[21,27]
14	$\left[\frac{1}{2}-x,\ \frac{1}{4},\ z\right]$	[22,28]
15	$\left[x,\ \frac{3}{4},\ z\right]$	[23,25]
16	$\left[x,\ \frac{1}{4},\ -z\right]$	[24,26]

Table 22: Wyckoff site: 16v, site symmetry: .m'.

No.	position	mapping
1	$[x,\ 0,\ z]$	[1,23]
2	$[x,\ 0,\ -z]$	[2,24]
3	$\left[\frac{1}{2}-x,\ 0,\ -z\right]$	[3,21]
4	$\left[\frac{1}{2}-x,\ 0,\ z\right]$	[4,22]
5	$[-x,\ 0,\ -z]$	[5,19]
6	$[-x,\ 0,\ z]$	[6,20]
7	$\left[x+\frac{1}{2},\ 0,\ z\right]$	[7,17]
8	$\left[x+\frac{1}{2},\ 0,\ -z\right]$	[8,18]
9	$\left[x+\frac{1}{2},\ \frac{1}{2},\ z\right]$	[9,31]
10	$\left[x+\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[10,32]
11	$\left[-x,\ \frac{1}{2},\ -z\right]$	[11,29]
12	$\left[-x,\ \frac{1}{2},\ z\right]$	[12,30]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ -z\right]$	[13,27]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ z\right]$	[14,28]
15	$\left[x,\ \frac{1}{2},\ z\right]$	[15,25]
16	$\left[x,\ \frac{1}{2},\ -z\right]$	[16,26]

Table 23: Wyckoff site: 16w, site symmetry: m..

No.	position	mapping
1	$[0,\ y,\ z]$	[1,6]
2	$[0,\ -y,\ -z]$	[2,5]
3	$\left[\frac{1}{2},\ y,\ -z\right]$	[3,8]
4	$\left[\frac{1}{2},\ -y,\ z\right]$	[4,7]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[9,14]

continued ...

Table 23

No.	position	mapping
6	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[10,13]
7	$\left[0,\ y+\frac{1}{2},\ -z\right]$	[11,16]
8	$\left[0,\ \frac{1}{2}-y,\ z\right]$	[12,15]
9	$\left[\frac{1}{2},\ y,\ z\right]$	[17,22]
10	$\left[\frac{1}{2},\ -y,\ -z\right]$	[18,21]
11	$[0,\ y,\ -z]$	[19,24]
12	$[0,\ -y,\ z]$	[20,23]
13	$\left[0,\ y+\frac{1}{2},\ z\right]$	[25,30]
14	$\left[0,\ \frac{1}{2}-y,\ -z\right]$	[26,29]
15	$\left[\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[27,32]
16	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	[28,31]

Table 24: Wyckoff site: 16x, site symmetry: m'..

No.	position	mapping
1	$\left[\frac{1}{4},\ y,\ z\right]$	[1,22]
2	$\left[\frac{1}{4},\ -y,\ -z\right]$	[2,21]
3	$\left[\frac{1}{4},\ y,\ -z\right]$	[3,24]
4	$\left[\frac{1}{4},\ -y,\ z\right]$	[4,23]
5	$\left[\frac{3}{4},\ -y,\ -z\right]$	[5,18]
6	$\left[\frac{3}{4},\ y,\ z\right]$	[6,17]
7	$\left[\frac{3}{4},\ -y,\ z\right]$	[7,20]
8	$\left[\frac{3}{4},\ y,\ -z\right]$	[8,19]
9	$\left[\frac{3}{4},\ y+\frac{1}{2},\ z\right]$	[9,30]
10	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ -z\right]$	[10,29]
11	$\left[\frac{3}{4},\ y+\frac{1}{2},\ -z\right]$	[11,32]
12	$\left[\frac{3}{4},\ \frac{1}{2}-y,\ z\right]$	[12,31]
13	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ -z\right]$	[13,26]
14	$\left[\frac{1}{4},\ y+\frac{1}{2},\ z\right]$	[14,25]
15	$\left[\frac{1}{4},\ \frac{1}{2}-y,\ z\right]$	[15,28]
16	$\left[\frac{1}{4},\ y+\frac{1}{2},\ -z\right]$	[16,27]

Table 25: Wyckoff site: 16y, site symmetry: ..m'

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,24]
2	$[x,\ -y,\ 0]$	[2,23]
3	$\left[\frac{1}{2}-x,\ y,\ 0\right]$	[3,22]
4	$\left[\frac{1}{2}-x,\ -y,\ 0\right]$	[4,21]
5	$[-x,\ -y,\ 0]$	[5,20]
6	$[-x,\ y,\ 0]$	[6,19]
7	$\left[x+\frac{1}{2},\ -y,\ 0\right]$	[7,18]
8	$\left[x+\frac{1}{2},\ y,\ 0\right]$	[8,17]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	[9,32]

continued ...

Table 25

No.	position	mapping
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[10,31]
11	$\left[-x,\ y+\frac{1}{2},\ 0\right]$	[11,30]
12	$\left[-x,\ \frac{1}{2}-y,\ 0\right]$	[12,29]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ 0\right]$	[13,28]
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	[14,27]
15	$\left[x,\ \frac{1}{2}-y,\ 0\right]$	[15,26]
16	$\left[x,\ y+\frac{1}{2},\ 0\right]$	[16,25]

Table 26: Wyckoff site: 16z, site symmetry: ..m'

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{2}\right]$	[1,24]
2	$\left[x,\ -y,\ \frac{1}{2}\right]$	[2,23]
3	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}\right]$	[3,22]
4	$\left[\frac{1}{2}-x,\ -y,\ \frac{1}{2}\right]$	[4,21]
5	$\left[-x,\ -y,\ \frac{1}{2}\right]$	[5,20]
6	$\left[-x,\ y,\ \frac{1}{2}\right]$	[6,19]
7	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}\right]$	[7,18]
8	$\left[x+\frac{1}{2},\ y,\ \frac{1}{2}\right]$	[8,17]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[9,32]
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[10,31]
11	$\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[11,30]
12	$\left[-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[12,29]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[13,28]
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[14,27]
15	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[15,26]
16	$\left[x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[16,25]

Table 27: Wyckoff site: 32A, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[x,\ -y,\ -z]$	[2]
3	$\left[\frac{1}{2}-x,\ y,\ -z\right]$	[3]
4	$\left[\frac{1}{2}-x,\ -y,\ z\right]$	[4]
5	$[-x,\ -y,\ -z]$	[5]
6	$[-x,\ y,\ z]$	[6]
7	$\left[x+\frac{1}{2},\ -y,\ z\right]$	[7]
8	$\left[x+\frac{1}{2},\ y,\ -z\right]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[9]
10	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[10]
11	$\left[-x,\ y+\frac{1}{2},\ -z\right]$	[11]
12	$\left[-x,\ \frac{1}{2}-y,\ z\right]$	[12]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$	[13]

continued ...

Table 27

No.	position	mapping
14	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[14]
15	$\left[x,\, \frac{1}{2}-y,\, z\right]$	[15]
16	$\left[x,\, y+\frac{1}{2},\, -z\right]$	[16]
17	$\left[x+\frac{1}{2},\, y,\, z\right]$	[17]
18	$\left[x+\frac{1}{2},\, -y,\, -z\right]$	[18]
19	$[-x,\, y,\, -z]$	[19]
20	$[-x,\, -y,\, z]$	[20]
21	$\left[\frac{1}{2}-x,\, -y,\, -z\right]$	[21]
22	$\left[\frac{1}{2}-x,\, y,\, z\right]$	[22]
23	$[x,\, -y,\, z]$	[23]
24	$[x,\, y,\, -z]$	[24]
25	$\left[x,\, y+\frac{1}{2},\, z\right]$	[25]
26	$\left[x,\, \frac{1}{2}-y,\, -z\right]$	[26]
27	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[27]
28	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[28]
29	$\left[-x,\, \frac{1}{2}-y,\, -z\right]$	[29]
30	$\left[-x,\, y+\frac{1}{2},\, z\right]$	[30]
31	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[31]
32	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[32]